

Exercise 1.7

29) Construct 3×2 matrices A and B such that $Ax=0$ has only the trivial solution and $Bx=0$ has a nontrivial solution.

Solution: Since $Ax=0$ has only the trivial solution, the columns of A are linearly independent. Since there are only 2 columns, neither of the columns is a multiple of the other.

Claim: $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 5 \end{bmatrix}$ is a valid matrix according to given conditions.

Proof: Let, $\begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} = c \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ for some $c \in \mathbb{R}$.

$$\therefore \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} c \\ 2c \\ 3c \end{bmatrix} \quad (\text{Scalar multiplication})$$

$$\therefore c=2, \quad 2c=1 \Rightarrow c=1/2.$$

This is a contradiction. $\therefore \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$ is ^{not a} scalar multiple of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Similarly, we can show that $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is not a scalar multiple of $\begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$. $\square$

• Since $Bx=0$ has a nontrivial solution, the columns of B are linearly dependent. Since there are only 2 columns, at least one column is a scalar multiple of the other.

• We know $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. $\therefore B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$ can be such a valid B . $\square$

a) 30) Fill in the blank in the following statement: "If A is an $m \times n$ matrix, then the columns of A are linearly independent if and only if A has n pivot columns".

b) Explain why the statement in (a) is true.

Solution: Since the columns of A are linearly independent, the matrix equation $Ax=0$ has a unique trivial solution. By Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$. By the row reduction algorithm and Theorem 1, the augmented matrix $[A \ 0]$ can be converted